Sofya Dymchenko, Ph.D.

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Work Experience

Doctoral Researcher, Grenoble, France — Mar 2022 – Apr 2026
Datamove team, INRIA and University of Grenoble-Alpes (UGA)

- Developed novel active learning methodology for deep PDE surrogates: created method for PINNs outperforming existing approaches (offline), pioneered extension to online setting for data-driven surrogates where training occurs concurrently with running simulations — improving training efficiency and reducing simulation costs.
- Extended open-source HPC&AI framework Melissa; supervised 2 engineers and 2 master students internships (Feb – Sep 2025) contributing to the project.

Research Intern, Moscow, Russia — Jun 2021 – Dec 2021
Computational Intelligence Lab, Skolkovo Institute of Science and Technology (Skoltech)

- Continued research on optimal transport algorithms from master's thesis; explored extensions to computational geometry applications and computer vision tasks.

Research Intern, Moscow, Russia — Feb 2020 – Aug 2020
Media Algorithms Laboratory, Huawei R&D Center

- Developed and trained generative models with attention mechanisms for audio processing and personalized noise reduction; contributed to R&D pipeline development in a product team environment.

Education

PhD in Computer Science, Grenoble, France — Mar 2022 – Apr 2026
Laboratory of Informatics of Grenoble (LIG), UGA

- Thesis: High Performance Online Deep Neural Network Training from Synthetic Data with Active Learning for Scientific Computing. The manuscript is rated "Outstanding" (top 5%) by external reviewer. [link].

MSc in Applied Mathematics and Computer Science, Moscow, Russia — Sep 2019 – Jun 2021
Statistical Learning Theory program, Skoltech jointly with HSE (with honors, GPA: 4.89/5)

- Thesis: Developed modified optimal transport algorithm reducing complexity from quadratic to log-linear, applied to computational mesh generation [link].

BSc in Mathematics, Moscow, Russia — Sep 2015 – Jun 2019
Faculty of Mathematics, HSE (GPA: 7.87/10); Minor in Data Science, Faculty of Computer Science, HSE

- Thesis: Collected and preprocessed 10-hours noisy YouTube dataset from scratch for rare languages multi-classification; trained multimodal CNN-RNN models achieving 77.6% accuracy [link].

Publications

- P. Cesar, **S. Dymchenko**, A. Purandare, and B. Raffin. Learning Where to Simulate: Generative Active Sampling for Online PDE Surrogate Training. ArXiv, June 2026 [preprint]
- **S. Dymchenko**, A. Purandare, and B. Raffin. MelissaDL x Breed: Towards Data-Efficient On-line Supervised Training of Multi-parametric Surrogates with Active Learning. In *SC-W 2024 - Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, Nov. 2024 [paper] [video]
- **S. Dymchenko** and B. Raffin. Loss-driven sampling within hard-to-learn areas for simulation-based neural network training. In *MLPS 2023 - Machine Learning and the Physical Sciences Workshop at NeurIPS 2023 - 37th Conference on Neural Information Processing Systems*, Dec. 2023 [paper]

Additional Experience & Skills

- Teaching & **leadership** (2019–2024): Led Python course (3 years, 100+ students); curriculum design, TA supervision, assessment, and lectures; "Best Course" award (2020). Delivered seminars in Data Science, Machine Learning, and Python to students from diverse backgrounds (BSc/MSc/professional) at HSE and UGA.
- Event **organization**: poster moderator, on-site support, outreach for Grenoble AI for Physical Sciences workshop [2024][2026]; local events (LIG).
- Academic & community **service**: reviewing for Workshop on Advancing Neural Network Training at NeurIPS23 and ICML24; engagement in doctoral school student council (UGA) and laboratory synergy initiatives (LIG); volunteering in animal shelter (SPA) and bird protection organization (LPO).

Technical skills: Python (PyTorch, JAX), Git, Bash, SLURM, Linux/HPC environments, C++ (basic), LaTeX.
Languages: English (fluent), French (intermediate), Russian (native).